Sagnick Mukherjee

51 Pegasi b Postdoctoral Fellow, School of Earth & Space Exploration, Arizona State University
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h-Index: 31 Citations: 4122 First-Author Papers: 13 Other Co-Authored Papers: 58
Software Developed: PICASO 3.0, VIRGA Students Advised: 14

Research Snapshot

My research asks a fundamental question: how do planets and substellar objects form and evolve, and what shapes their observed diversity today? I address this by developing theoretical frameworks for understanding their atmospheres across a broad mass range, from brown dwarfs and giant exoplanets to sub-Neptunes and terrestrial worlds. I am a leader in exoplanet atmosphere and interior theory, and have built state-of-the-art models, including PICASO and the Sonora atmospheric and evolutionary model grids, to connect atmospheric spectra with the physical conditions, interiors, and long-term evolution of exoplanets and brown dwarfs. I also have expertise in observational studies of exoplanet atmospheres, using optical/infrared spectroscopy from space-based observatories such as JWST and ground-based facilities such as Gemini. By combining theory, modeling, and observations, I seek to unravel the physical, chemical, and geological processes that govern these worlds, reveal the mechanisms that shape planetary origins and diversity, and assess their potential for habitability.

Research Interests

1. Theory of exoplanet & brown dwarf atmospheres.
2. Interior and Evolution of substellar objects.
3. Characterization of exoplanet atmospheres with the James Webb Space Telescope, Hubble Space Telescope, and ground-based observatories.
4. Open-source software development in astronomy.

Education

Ph.D., Astronomy & Astrophysics Oct 2020 – Aug 2025
University of California, Santa Cruz; Advisor: Prof. Jonathan Fortney.
Thesis: *Understanding Atmospheres of Exoplanets and Brown Dwarfs with JWST.*

M.S., Astronomy & Astrophysics Oct 2020 – Dec 2022
University of California, Santa Cruz; GPA: **4.0/4.0**.

M.Sc., Physics Sep 2018 – Jul 2020
Presidency University, Kolkata; GPA: **9.65/10**.

B.Sc., Physics Aug 2015 – Sep 2018
Presidency University, Kolkata; GPA: **9.21/10** (**Gold Medalist**).

Appointments

51 Pegasi b Postdoctoral Fellow Sep 2025 – present
School of Earth & Space Exploration, Arizona State University (SESE).
Advisors: Prof. Luis Welbanks & Prof. Michael Line

Templeton Theory-Experiment Cross Training Fellow June 2023 – August 2025
Dept. of Physics & Astronomy, Johns Hopkins University.
Advisor: Prof. David Sing

S N Bose Summer Research Fellow June 2019 – August 2019
Dept. of Astronomy & Astrophysics, UC Santa Cruz.
Advisor: Prof. Puragra Guhathakurta

Awards and Fellowships

1. **51 Pegasi b Postdoctoral Fellowship (2025)** — Heising–Simons Foundation, host: SESE, ASU.
2. **NASA Hubble Postdoctoral Fellowship (2025-2028), Sagan Fellowship** — hosted at University of Texas, Austin (**Declined**).
3. **Templeton Theory–Experiment Cross Training (TEX) Fellowship** — co-hosted by **UCSC** and **JHU**.
4. **UC President's Lindau Fellow 2024** — Participant, **73rd Lindau Nobel Laureate Meeting (Physics)**, Lindau, Germany.
5. **Barbara Walker Best Paper Award (2023–24)** for "*PICASO 3.0: A One-dimensional Climate Model for Giant Planets and Brown Dwarfs*".
6. **UC Regents Fellowship** — first-year graduate study, Dept. of Astronomy & Astrophysics, UCSC.
7. **Whitford Prize** — highest achievement in research, coursework, and teaching as a 2nd-year Graduate Student, UCSC Astronomy.
8. **S.N. Bose Scholarship (IUSSTF, 2019)** — short-term summer research at UC Santa Cruz.
9. **Gold Medalist — M.Sc. Physics** (2018–2020) and **B.Sc. Physics** (2015–18).
10. **Jagadis Bose National Science Talent Search Scholarship (2015–2020)** — undergraduate research.

Highlighted Publications (full list at the end)

5 most cited or impactful first-authored publications

- "Cloudy Mornings and Clear Evenings on a Gas Giant Exoplanet", **Sagnick Mukherjee**, David Sing, *et al.*, (2026b, **Science**)
- "The Sonora Substellar Atmosphere Models. IV. Elf Owl: Atmospheric Mixing and Chemical Disequilibrium With Varying Metallicity and C/O Ratios", **Sagnick Mukherjee**, Jonathan J. Fortney, *et al.*, (2024, **The Astrophysical Journal**)
- "PICASO 3.0: A One-Dimensional Climate Model for Giant Planets and Brown Dwarfs", **Sagnick Mukherjee**, Natasha E. Batalha, *et al.*, (2023, **The Astrophysical Journal**)

- "Probing the Extent of Vertical Mixing in Brown Dwarf Atmospheres with Disequilibrium Chemistry", **Sagnick Mukherjee**, Jonathan J. Fortney, *et al.,* (2022, **The Astrophysical Journal**)
- "Effects of Planetary Parameters on Disequilibrium Chemistry in Irradiated Planetary Atmospheres: From Gas Giants to Sub-Neptunes", **Sagnick Mukherjee**, Jonathan J. Fortney, *et al.*, (2025, **The Astrophysical Journal**)

Grants & Observing Proposal Awards

Grant Summary

JWST AR Theory Programs & GO Observing Programs $648,797
Role: 3 Principal Investigator, 2 Co-Investigator

Templeton TEX Fellowship $234,462
Role: Principal Investigator

51 Pegasi b Fellowship $436,025
Role: Principal Investigator

Observing Proposal Awards

Approved PI programs

1. **Co-PI** of Awarded **James Webb Space Telescope (JWST)** large **Cycle 5 GO 9709** "Parting the Clouds: An optimized limb-limb exoplanet survey to reveal diurnal cloud cycling and the mass-metalicity relation"
2. **PI** of Awarded **James Webb Space Telescope (JWST) Cycle 4 AR GO 7358** "The Other Extreme: Enabling Characterization of Metal-poor Brown Dwarfs and testing our Understanding of Jupiter and Saturn"
3. **Co-PI** and **theory lead** of Awarded **JWST Cycle 2** time for the **GO 4094** "Probing the Depths: Disequilibrium Chemistry as a Tracer of Mixing Processes in Brown Dwarf Atmospheres".
4. **PI** of Awarded **JWST Cycle 2 AR GO 3245** "Up to the Task? A New Generation of Atmospheric and Interior Models of Brown Dwarfs for the JWST Era".
5. **PI** of Awarded **Gemini South** time (2026A-188207) for the program "Chasing Clouds from Dawn to Dusk: Probing Aerosol Cycling in Hot Jupiters".

Approved co-I programs

6. **Co-I** of Awarded **Gemini North** time (2026A-FT-201) for the program "Leveraging IGRINS2 observations to resolve opposing JWST interpretations on a dense Saturn mass exoplanet and its formation history".
7. **Co-I** of Awarded **Devasthal Optical Telescope** time for the program (DOT-2026-C1-P28) "Living on the Edge: Deciphering the Evolutionary History of Transitionary Exoplanets and Brown Dwarf".
8. **Co-I** of Awarded **Devasthal Optical Telescope** time for the program (DOT-2026-C1-P48) "Study of architecture of exoplanetary systems".
9. **Co-I** of Awarded **JWST Cycle 5 GO 11831** "FOURCAST: FOUR-planet Comparative AtmoSpheric Escape in a benchmark young multi-planet system"

10. **Co-I** of Awarded **JWST Cycle 5 GO 11144** "Unveiling the Mysteries of the warm Neptune GJ 436b: Probing the atmospheric composition and interior properties"

11. **Co-I** of Awarded **JWST Cycle 5 GO 9942** "Searching for the Fire Within: Investigating the Impact of Tidal Heating on Exoplanet Interiors and Atmospheres with JWST"

12. **Co-I** of Awarded **JWST Cycle 4 GO 9025** "The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution"

13. **Co-I** of Awarded **JWST Cycle 4 GO 8004** "Cliff Hangers: Testing for Atmosphere-Mantle Interactions in Radius Cliff Planets".

14. **Co-I** of Awarded **JWST Cycle 4 GO 9095** "Combining Emission and Transmission Spectroscopy to reveal Exo-Neptune Aerosols, Chemistry, and Formation".

15. **Co-I** of Awarded **JWST Cycle 4 GO 8140** "Empirically anchoring the physics of silicate clouds using L0- T9 benchmark brown dwarfs".

16. **Co-I** of Awarded **JWST Cycle 3 Large GO 5959** "KRONOS: Keys to Revealing the Origin and Nature Of sub-neptune Systems".

17. **Co-I** of Awarded **JWST Cycle 3 GO 6122** "Cool kids on the block: The direct detection of cold ice giants and gas giants orbiting young low-mass neighbors".

18. **Co-I** of Awarded **JWST Cycle 2 GO 3231** "Panchromatic Phase Curve of the Highest-S/N Hot Neptune".

19. **Co-I** of Awarded **JWST Cycle 2 AR GO 3201** "The Utility of Self-Consistent Models and Photochemistry in Understanding Transiting Planet Atmospheres".

20. **Co-I** of Awarded **JWST Cycle 2 DD 4558** "Establishing the Formation of AF Lep b with NIRCam: The Lowest-Mass Imaged Exoplanet with a Dynamical Mass".

21. **Member** of the **JWST grand tour collaboration** from the awarded **JWST Cycle 3 GO 5924** large program.

22. **Member** of the **JWST Morning/Evening collaboration** from the awarded **JWST Cycle 2 GO 3969** program.

23. **Member** of the **JWST MANATEE GTO collaboration**.

24. **Co-I** of Awarded **uGMRT Cycle 47** GO program "Investigating the Auroral Heating of Brown Dwarf Atmosphere with uGMRT and JWST".

25. **Co-I** of Awarded **uGMRT Cycle 47** GO program "Can rogue super-Jupiters be radio-bright? A search for radio emission at the deuterium burning limit".

26. **Co-I** of Awarded **HST Cycle 30 GO 17136** "Photometry of a Young Planetary-Mass Companion to a Taurus M Dwarf Star".

27. **Co-I** of Awarded **HST Cycle 30 AR 17064** "Advancing our understanding of stellar variability with the Panchromatic Hubble Andromeda Treasury".

Teaching & Mentoring

Advised 4 graduate, 4 undergraduate, & 6 high-school students, Taught 2 courses

1. **Co-advising** PhD student Arjun Dawn (ARIES, India) for the thesis project "Studying Exoplanetary Atmospheres in High Resolution with the DOT-HRS instrument" with Prof. Jeewan Pandey, 2025-present
2. **Co-advising** graduate student Finnegan Keller (SESE, ASU) for the project "Statistical Techniques to Extract Limb Resolved Spectra of Giant Exoplanets from JWST data" with Prof. Luis Welbanks, 2025-present.
3. **Mentored** PhD student Aditya Sengupta (UCSC) for the project "Implementation of user defined aerosol properties within PICASO", 2024.
4. **Mentored** PhD student Anna Gagnebin (UCSC) for the project "Simulating the Climates of Volatile Rich sub-Neptunes", 2024.
5. **Mentoring** Master's students Saibal Dey Biswas and Tiyasa Pyne (Presidency University, Kolkata, India) for their Master's thesis project "Limb Asymmetries in Giant Exoplanets", 2025-present.
6. **Mentored** undergraduate student Arya Jhamb (UCSC) for the project "Retrievals on Brown Dwarf Spectra in the JWST Era", 2024
7. **Mentored** undergraduate student Anna Gagnebin (California State University, Sacramento) for the project "Exoplanet Atmosphere Models for JWST Spectroscopy", 2022.
8. **Mentored 6 high school students** for the astronomy project "Photometrically variable stars in M31" as a part of the **Science Internship Program (SIP) 2019 and 2020**.
9. **Teaching Assistant** for **ASTR-10** "From the Big Bang to Planet Earth" with Prof. Alexie Leauthaud, Spring 2023.
10. **Teaching Assistant** for **ASTR-16** "Astrobiology: Life in the Universe" with Prof. Natalie Batalha, Fall 2021.

Colloquia and Seminars

1. "Cloudy Mornings and Clear Evenings in Giant Exoplanets", **September 2026**, **SESE, Arizona State University**, Tempe.
2. "Deciphering Exoplanets: From JWST Breakthroughs to a Global Census", **February 2026**, **Institute for Astronomy, University of Hawai'i, Manoa**.
3. "Exoplanet Atmospheres in 3D: A New Lens on Physics and Chemistry", **February 2026**, **Institute for Astronomy, University of Hawai'i, Manoa**.
4. "Exoplanet Atmospheres in 3D: A New Lens on Physics and Chemistry", **March 2026**, **Dept. of Physics & Astronomy, Clemson University**.
5. "Deciphering Exoplanets: From JWST breakthroughs to a global census", **February 2026**, **Dept. of Astronomy & Astrophysics, Raman Research Institute**, Bengaluru.

6. "Constraints on Atmospheric Mixing in Brown Dwarf and Transiting Exoplanet Atmospheres in the JWST Era", **September 2024**, **Dept. of Astronomy, UT Austin**, Texas.

7. "Constraints on Atmospheric Mixing in Transiting Exoplanet and Brown dwarf Atmospheres in the JWST Era", **September 2024**, **Lunar and Planetary Laboratory, Univ. of Arizona**, Tucson.

8. "Constraints on Atmospheric Mixing in Brown Dwarf and Exoplanet Atmospheres in the JWST Era", **February 2024**, **Carnegie Earth and Planets Laboratory**, Washington DC.

9. "Constraints on Atmospheric Mixing in Brown Dwarf and Exoplanet Atmospheres in the JWST Era", **November 2023**, **CTC Seminar, Univ. of Maryland**, College Park.

10. "Constraints on Atmospheric Mixing in Brown Dwarf and Exoplanet Atmospheres in the JWST Era", **November 2023**, **Dept. of Astronomy, Cornell University**, Ithaca.

11. "Constraints on Atmospheric Mixing in Brown Dwarf and Exoplanet Atmospheres in the JWST Era", **October 2023**, **School of Earth and Planetary Sciences, NISER**, Bhubaneswar, India.

12. "Constraints on Atmospheric Mixing in Brown Dwarf and Exoplanet Atmospheres in the JWST Era", **November 2023**, **Dept. of Astrophysics and High Energy Physics, SNBNCBS**, Kolkata, India.

13. "Constraints on Atmospheric Mixing in Brown Dwarf and Exoplanet Atmospheres in the JWST Era", **July 2023**, **Imperial College London**, London.

14. "Constraining Vertical Mixing, Metallicity, and C/O Ratio of Giant Planet and Brown Dwarf Atmospheres", **March 2023**, **SESE, Arizona State University**, Tempe.

15. "Understanding Exoplanet and Brown Dwarf Atmospheres in the JWST Era", **June 2022**, **CESSI Seminar, IISER Kolkata**.

Conferences, Talks, & Posters

1. **Invited Review Talk**, "Self-consistent atmospheric modeling of substellar objects", **July, 2026, Charting Substellar Worlds: Atmospheres, Formation, and Future Tools for the Next Decade of Discovery**, Monterey, California.

2. **Contributed Talk**, "Cloudy mornings and clear evenings on gas giant exoplanets", **August 2026**, **Other Worlds Laboratory Workshop**, Santa Cruz, California.

3. **Contributed Talk**, "Cloudy morning and clear evening on a gas giant exoplanet", **March 2026**, **Exoplanet Atmospheres**, Denver, Colorado.

4. **Contributed Talk**, "A Panchromatic and Physically Consistent View of K2-18 b's Atmosphere ", **January 2026**, **247th AAS**, Phoenix, AZ.

5. **Dissertation Presentation**, "Disequilibrium Chemistry in Planet Atmospheres: from Brown Dwarfs to sub-Neptunes", **January 2025**, **245th AAS**, National Mall, Washington DC

6. **Contributed Talk**, "Cloudy Mornings and Clear Evenings in a Giant Extrasolar World", **August 2025**, **51 Pegasi b Summit**, Half Moon Bay, California

7. **Contributed Talk**, "Effects of Planetary Parameters on Disequilibrium Chemistry: From Gas Giants to Sub-Neptunes", **July 2024**, **MPIA Heidelberg** — Challenge Accepted! Linking Planet Formation with Present-Day Atmospheres.

8. **Contributed Talk**, "Revisiting GJ 436b's Atmosphere with Panchromatic JWST Emission Spectroscopy", **June 2024**, **Exoplanets V**, Leiden, Netherlands.

9. **Contributed Talk**, "PICASO: An Unified Atmospheric Model of Exoplanetary Atmospheres with Photochemistry and Vertical Mixing", **June 2023**, **ERES Conference**, Yale University.

10. **Poster Presentation**, "Constraints on Atmospheric Vertical Mixing in Giant Exoplanets and Brown Dwarfs", **September 2023**, **Exoplanets: Atmospheres to Architecture**, Washington DC.

11. **Poster Presentation**, "PICASO: An Unified Atmospheric Model of Exoplanetary Atmospheres with Photochemistry and Vertical Mixing", **June 2023**, **Exoclimes Conference**, University of Exeter.

12. **Contributed Talk**, "Atmospheric and Evolutionary Models of Substellar Objects with Disequilibrium Chemistry for the JWST Era", **January 2023**, **AAS Conference**, Seattle.

13. **Contributed Talk**, 'Atmospheric and Evolutionary Models of Substellar Objects with Disequilibrium Chemistry for the JWST Era', **January 2023**, **ExoPAG 27**, Seattle.

14. **Contributed Talk**, 'PICASO 3.0: A One-Dimensional Open Source Climate Model for Giant Planets and Brown Dwarfs', **October 2022**, **42nd Bay Area Exoplanets Meeting**, SETI.

15. **Contributed Talk**, 'Understanding Atmospheric Mixing with Disequilibrium Chemistry in Brown Dwarfs', **July 2022**, **Other Worlds Laboratory (OWL) Summer Program**.

16. **Poster Presentation**, 'Measuring Vertical Mixing in Giant Planets and Brown Dwarf Atmospheres', **September 2023**, **Giant Magellan Telescope Community Science Meeting**, Washington DC.

17. **Poster Presentation**, 'PICASO+VULCAN: Modeling Exoplanetary Atmospheres Self-Consistently with Photochemistry and Vertical Mixing', **June 2023**, **Exoclimes VI**, University of Exeter, UK.

18. **Poster Presentation**, 'Probing Atmospheric Mixing with Disequilibrium Chemistry in Brown Dwarfs and Warm Exoplanets', **May 2022**, **Exoplanets IV**, Las Vegas.

19. **Contributed Talk**, 'Understanding Atmospheric Mixing with Disequilibrium Chemistry in Brown Dwarfs', **January 2022**, **CHAMPs Exoplanet Early Career Seminar**.

20. **Contributed Talk**, 'Modeling Polarization Signals from Cloudy Brown Dwarfs: Luhman 16 A and B in Three Dimensions', **September 2021**, **Bay Area Exoplanet Meeting 38**.

21. **Poster Presentation**, 'Modeling Polarization signals in 3D from brown dwarfs Luhman 16 and B', **April 2021**, **STScI Spring Symposium**.

22. **Contributed Talk**, 'Cloud Complexity Required for Retrievals on Reflected Spectroscopy of Cool Giants', **September 2020**, **Bay Area Exoplanet Meeting 34**.

23. **Contributed Talk**, 'The accretion disc-jet connection in blazars', **Spring 2019**, **37th ASI Annual Meeting**, Christ University, Bangalore.

Open-Source Software & Scientific Computing

1. **Developed** the open-source atmospheric model **PICASO 3.0** for exoplanets and brown dwarfs.

2. **Developed** the **Sonora** interior and evolution model for exoplanets and brown dwarfs.

3. **Contributed** to development of exoplanet cloud modeling package **VIRGA**.
4. **Contributed** to development of the **AURORA** atmospheric inference model.
5. **7 years** experience with exoplanet atmospheric simulation package **PICASO**.
6. Proficient scientific programming with **Python** and **Fortran**.
7. Proficient in **GPU**-based Python programming with **Numba CUDA** and **CuPy**.
8. **Selected and Participated** in the **NASA GPU Hackathon 2022** co-organized by **NASA** and **NVIDIA**.

Professional Service

1. **Referee** for leading astronomy journals including *The Astrophysical Journal* and *Astronomy & Astrophysics*
2. Served as a **reviewer** for scientific funding proposal panels for NASA.
3. Served as a **trainer** for the **PICASO** software in the **Sagan Summer Workshop, 2023**.
4. Served as a **mentor** for the **PICASO** hands-on session in the **Sagan Summer Workshop, 2021**.
5. Co-designer of "Atmospheric modelling and retrieval" workshop in the upcoming Roman Coronagraph School Europe 2026 to be held in France, March 2026.
6. Organizer of the Exoplanet and Brown Dwarf Journal Club at SESE, ASU.

Media & Press

1. "JWST sees partly cloudy skies on a distant, giant exoplanet ", **Scientific American**, May 21, 2026.
2. "See the clouds streaming and vanishing around this planet — 690 light years away", **Nature**, May 21, 2026.
3. "Webb Telescope Detects Cloudy Mornings And Clear Nights On Alien World", **Forbes**, May 21, 2026.
4. "Cloudy mornings, clear nights: charting weather on distant world", **The Times**, May 21, 2026.
5. "How Clouds Hide Alien Worlds—and How Astronomers See Through Them?", **SETI live podcast**, June 18, 2026
6. "JWST maps the weather on a hot gas giant 700 light-years away", **Ars Technica**, May 21, 2026.
7. "Astronomers de-fog exoplanet atmospheres with new cloud-detecting method", **Phys.org**, May 21, 2026.
8. "Astronomers de-fog exoplanet atmospheres with new cloud-detecting method", **Johns Hopkins University Newscenter**, May 21, 2026.
9. "Astronomers get a peek at exoplanet atmospheres with new cloud-detecting method", **ASU News**, May 21, 2026.

10. "Methane Throughout The Atmosphere Of The Warm Exoplanet WASP-80b", **Astrobiology**, September 7, 2023.
11. "Astronomy PhD candidate researching mysteries of sub-Neptune planets wins fellowship", **UCSC Newscenter**, March, 2025.
12. "JWST makes first unequivocal detection of carbon dioxide in an exoplanet atmosphere", **UCSC Newscenter**, August 25, 2022.
13. "James Webb telescope detects dust storm on distant world", **BBC News**, March 22, 2023.
14. "NASA's Webb Detects Carbon Dioxide in Exoplanet Atmosphere", **NASA JPL**, August 25, 2022. Also covered by leading news platforms across the world.
15. "JWST's First Direct Spectrum of a Planetary-Mass Object", **Astrobites**, September 2, 2022.
16. "NASA releases Webb telescope's first exoplanet image", **UCSC Newscenter**, September 1, 2022. Also covered by leading news platforms across the world.

Publications

Papers published or in review in leading journals including *Science*, *Nature*, *Nature Astronomy*, *PNAS*, *ApJ*, *ApJL*, *MNRAS*, *ApJS*, and *AJ*.

Full NASA ADS library

Publications – First-Authored

12 published, 1 in review

1. "Cloudy Mornings and Clear Evenings on a Gas Giant Exoplanet", **Sagnick Mukherjee**, David Sing, *et al.*, (**2026b**, *Science*)
2. "Impact of Clouds on the Atmosphere-Mantle Interface of Sub-Neptunes", **Sagnick Mukherjee**, Matthew C. Nixon, *et al.*, (**2026c**, *The Astrophysical Journal Letters*)
3. "A New Method for Constraining 3D Exoplanet Atmospheres: Application to WASP-94A b's Panchromatic Spectrum", **Sagnick Mukherjee**, Luis Welbanks, *et al.*, (**2026d**, *under review in AAS journals*)
4. "The Impact of Irradiation on the Radius and Thermal Evolution of Transiting Brown Dwarfs", **Sagnick Mukherjee**, Jonathan Fortney, *et al.*, (**2026a**, The Astrophysical Journal)
5. "Effects of Planetary Parameters on Disequilibrium Chemistry in Irradiated Planetary Atmospheres: From Gas Giants to Sub-Neptunes", **Sagnick Mukherjee**, Jonathan J. Fortney, *et al.*, (**2025**, The Astrophysical Journal)
6. "A JWST Panchromatic Thermal Emission Spectrum of the Warm Neptune Archetype GJ 436b", **Sagnick Mukherjee**, Everett Schlawin, *et al.*, (**2025**, The Astrophysical Journal Letters (ApJL))
7. "The Sonora Substellar Atmosphere Models. IV. Elf Owl: Atmospheric Mixing and Chemical Disequilibrium With Varying Metallicity and C/O Ratios", **Sagnick Mukherjee**, Jonathan J. Fortney, *et al.*, (**2024**, The Astrophysical Journal)

8. "Probing the Extent of Vertical Mixing in Brown Dwarf Atmospheres with Disequilibrium Chemistry", **Sagnick Mukherjee**, Jonathan J. Fortney, *et al.,* (**2022**, The Astrophysical Journal (ApJ))

9. "PICASO 3.0: A One-Dimensional Climate Model for Giant Planets and Brown Dwarfs", **Sagnick Mukherjee**, Natasha E. Batalha, *et al.,* (**2023**, The Astrophysical Journal)

10. "Modeling Polarization Signals from Cloudy Brown Dwarfs Luhman 16 A and B in Three Dimensions", **Sagnick Mukherjee**, Jonathan J. Fortney, *et al.,* (**2021**, The Astrophysical Journal)

11. "Cloud Parameterizations and their Effect on Retrievals of Exoplanet Reflection Spectroscopy", **Sagnick Mukherjee**, Natasha E. Batalha, and Mark S. Marley (**2021**,The Astrophysical Journal)

12. "The accretion disc-jet connection in blazars", **Sagnick Mukherjee**, Kaustav Mitra, and Ritaban Chatterjee (**2019**, Monthly Notices of the Royal Astronomical Society)

13. "X-Ray Surface Brightness Profiles of Optically Selected Active Galactic Nuclei: Comparison with X-Ray AGNs", **Sagnick Mukherjee**, Anirban Bhattacharjee, *et al.,* (**2019**, The Astrophysical Journal)

Publications — 2nd & 3rd-Authored

12 published, 1 student-led, 1 under review

1. "Overcast Mornings and Clear Evenings in Hot Jupiter Exoplanet Atmospheres", Guangwei Fu, **Sagnick Mukherjee**, *et al.,* (**2025**, ApJL)

2. "Asymmetric Aerosol Distribution on the Terminators of the Warm Saturn WASP-69 b Revealed by JWST NIRISS/SOSS", Le-Chris Wang, **Sagnick Mukherjee**, *et al.,* (**2026**, under review in ApJL)

3. "JWST color sequence of transiting exoplanets and directly imaged substellar objects", Guangwei Fu, Joshua Lothringer, **Sagnick Mukherjee**, *et al.,*(**2026**, AJ)

4. "A Metal-poor Atmosphere with a Hot Interior for a Young Sub-Neptune Progenitor: JWST/NIRSpec Transmission Spectrum of V1298 Tau b", Saugata Barat, Jean-Michel Désert, **Sagnick Mukherjee**, *et al.,* (**2025**, AJ)

5. "Disequilibrium Chemistry, Diabatic Thermal Structure, and Clouds in the Atmosphere of COCONUTS-2b", Zhoujian Zhang, **Sagnick Mukherjee**, *et al.,* (**2024**, AJ)

6. "Bayesian Model Comparison and Significance: Widespread Errors and how to Correct Them", Daniel P. Thorngren, David K. Sing, **Sagnick Mukherjee** (**2025**, ApJS)

7. "A Tale of Two Molecules: The Underprediction of CO 2 and Overprediction of PH 3 in Late T and Y Dwarf Atmospheric Models", Samuel A. Beiler, **Sagnick Mukherjee**, *et al.,* (**2024**, ApJ)

8. "The atmosphere of HD 149026b: Low metal-enrichment and weak energy transport", Anna Gagnebin, **Sagnick Mukherjee**, *et al.,* (**2024**, ApJ); student-led

9. "Multiple Clues for Dayside Aerosols and Temperature Gradients in WASP-69 b from a Panchromatic JWST Emission Spectrum" Everett Schlawin, **Sagnick Mukherjee**, *et al.,* (**2024**, AJ)

10. "High-Precision Atmospheric Constraints for a Cool T Dwarf from JWST Spectroscopy", Callie E. Hood, **Sagnick Mukherjee**, *et al.,* (**2024**, Nature Astronomy, Under Review)

11. "The Sonora Substellar Atmosphere Models. III. Diamondback: Atmospheric Properties, Spectra, and Evolution for Warm Cloudy Substellar Objects" Caroline V. Morley, **Sagnick Mukherjee**, *et al.*, (**2024**, ApJ)

12. "Early Release Science of the exoplanet WASP-39b with JWST NIRSpec PRISM", Z. Rustamkulov, D. K. Sing, **Sagnick Mukherjee**, *et al.*, (**2023**, Nature)

13. "The Infrared Colors of 51 Eridani b: Micrometeoroid Dust or Chemical Disequilibrium?", Alexander Madurowicz, **Sagnick Mukherjee**, *et al.*, (**2022**, AJ)

Publications — Other Co-Authored

38 published, 7 in review

Published

1. "The Challenges of Detecting Gases in Exoplanet Atmospheres", Welbanks *et al.*, including **Sagnick Mukherjee** (**2025**, Nature Astronomy)

2. "A Clearer View of HAT-P-1 b: JWST NIRSpec G395H Reveals Water, Carbon Dioxide, and Possibly Hydrogen Sulfide", Ashtari *et al.*, including **Sagnick Mukherjee** (**2026**, AJ)

3. "A JWST transiting survey of FGK stellar limb darkening: empirical evidence for quadratic laws and atmospheric model comparisons", Sing *et al.*, including **Sagnick Mukherjee** (**2026**, AJ)

4. "PICASO 4.0: Clouds and Photochemistry in Climate Models of Brown Dwarfs and Exoplanets", Mang *et al.*, including **Sagnick Mukherjee** (**2026**, ApJ)

5. "The Sonora Substellar Atmosphere Models. VII. Flame Skimmer: Cloud-free Atmospheric and Evolutionary Models for the Coldest Substellar Objects", Mang *et al.*, including **Sagnick Mukherjee** (**2026**, ApJ)

6. "The Sonora Substellar Atmosphere Models VI. Red Diamondback: Extending Diamondback with SPHINX for Brown Dwarf Early Evolution", Davis *et al.*, including **Sagnick Mukherjee** (**2025**, ApJ)

7. "A Panchromatic Characterization of the Evening and Morning Atmosphere of WASP-107 b: Composition and Cloud Variations, and Insight into the Effect of Stellar Contamination", Murphy *et al.*, including **Sagnick Mukherjee** (**2025**, ApJ)

8. A Precise Metallicity and Carbon-to-Oxygen Ratio for a Warm Giant Exoplanet from its Panchromatic JWST Emission Spectrum Wiser *et al.*, including **Sagnick Mukherjee** (**2025**, PNAS)

9. "Atmospheric abundances and bulk properties of the binary brown dwarf Gliese 229 Bab from JWST/MIRI spectroscopy", Xuan *et al.*, including **Sagnick Mukherjee** (**2024**, ApJL)

10. "Statistical trends in JWST transiting exoplanet atmospheres", Fu *et al.*, including **Sagnick Mukherjee** (**2025**, ApJ)

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